

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8386**

AVI	Session:	Date:
2025-26	Odd MSE October 2025	08-10-2025
Sem:	Slot:	C. Code:
1	10:00 am - 11:00 am	CS413
Exam Type:	Block:3DS	Seat No:
Written	Course Section:1	2022700031

L_41_643_986_L_1505_41362



Name of the candidate. Mehunali Mahendrea More

Candidate's UID number:

2	0	2	2	7	0	0	0	3	1
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Examination class room number:

3	0	5
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Invigilator's signature with date:

 8/10
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(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: CSE - DS

I / II / III / IV / V / VI / VII / VIII ✓

DAY AND DATE: Wednesday, 08/10/25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.4.a ~~Q.4~~ $f(w) = \frac{1}{2} \|Xw - y\|^2$

To ~~prove and~~ derive: Derive gradient update rule.

To prove: The convergence requires the learning rate $\eta < \frac{2}{\lambda_{\max}}$

$\lambda_{\max} \rightarrow$ largest eigen value of $X^T X$

Weight update equations.

$$\Delta w = -\eta (\nabla f(w))$$

$$w_t = w_{t-1} + \Delta w_t$$

By using weight update equations gradient decent is evaluate.
For more regressive approach stochastic gradient decent is used.

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Question
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	Q1.b	Bias of a model and variance of a model determines how the model fits on the given dataset. Low bias and high variance leads to underfitting High bias and low variance leads to overfitting Capacity of a model refers to how well a model gets trained on a dataset and how much load it can manage and give correctly learn the model. If the capacity of a model is too high. it implies that the model will work well for larger datasets as well and the parameters & attributes can be larger in number hence there is a higher chance of the model getting overfitted. Thus, overfitting occurs when model capacity is too high.. This affects generalization as well.

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Q.2a

Input size = $H \times W \times C$
filter size = $F \times F$

Output size = $\frac{n - F + 2p}{S} + 1$

where, F = filter size

S = stride

P = padding

n = no. of inputs.

Computational complexity =

Depthwise separable convolutions are critical in modern architecture as they tend to give better results. Reconstructing and output becomes more feasible. It can efficiently utilize the parameters and use them to give better results.



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Q2b]

Autoencoders are unsupervised technique to learn of neural network. It has two components encoders that maps the inputs with ^{latent} ~~lowest~~ ^{representations} dimensions & decoder derives output from these representations.

There exists two broad categories of Autoencoders i.e. Undercomplete autoencoders & overcomplete autoencoders. Undercomplete autoencoders have mapping with size lower than the input, if the undercomplete autoencoder is trained with MSE it learns a subpace equivalent to PCA i.e. Principal Component analysis that is a dimensionality reduction technique.

This can be commented on the basis that Undercomplete Autoencoders maps with lower size dimensions which leads to dimensionality reduction just like that happens in PCA. Hence, we can say that Undercomplete autoencoders trained with MSE mimics PCA.

Overcomplete autoencoders have to map with dimensions of size greater than the input hence there are chances of overfitting.

To avoid this regularization is introduced. There are 3 types of regularized autoencoders - Sparse, denoising & convolutional autoencoders.

But overcomplete autoencoder without regularization tends to learn identity fn. because without regularization it overfits the model.

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Q.3a] Input Forget gate in LSTM helps in skipping on the unimportant previous information & compute the training in a more efficient manner. LSTMs have input gate, output gate & a forget gate.

In traditional RNNs the model is trained over previous data which might lead to less accurate results, the use of forget gates leads to more accurate results as unimportant details are skipped by the forget gate.

The forget gates modifies the recursion and prevents vanishing gradients under suitable initialization, this implies that recursive use of the forget gate leads to compute only from the parameters or gradients that have significance in further computations & not from the ones that are the vanishing gradients.

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Q.3b] GRUs have only update gate (which is a combination of input & forget gate) and an output gate whereas LSTM has input, output as well as forget gate. LSTMs have separate hidden state & cell state, whereas in GRUs cell states are merged / combined with hidden states.

~~Some~~ Fewer parameters in GRUs make them faster than LSTM, at places where fast computation is required GRUs can be used but at instances where accuracy matters LSTM is favoured.

At more generalized applications where accuracy does not matter GRUs are used.

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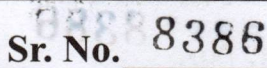


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